CocoaCheck Sabah — Latest Project Context

Oct 4, 2026 · @Sebastian

Status at a glance

CocoaCheck checks **whole dried cocoa beans**, not cocoa pods. The pod idea from earlier was a mix-up and is no longer part of the project. Some parts are still **working assumptions** until the trainer confirms them.

Labels used in this doc:

- **Decided**: the team has agreed
- **Working assumption**: current plan, waiting for the trainer's answer
- **Open**: not decided yet

Topic	Current position	Status
What the AI looks at	Whole dried cocoa beans (not pods, not cut beans)	Decided
What the AI says	Normal or non-normal, with how confident it is	Working assumption
Where the AI learns from	An existing public set of whole-bean photos	Working assumption
Photos per assessment	One bean photo as a spot check for the batch	Working assumption
Cut-test AI	Future idea, not in the first version	Decided
Login and admin	Not in the first version; maybe later	Decided
Quality rules	Demo values for now, clearly labelled, until LKM gives real ones	Working assumption
Demo Day	Live demo with one normal and one non-normal batch	Decided
LKM interview	Not held yet	Open

The project

CocoaCheck Sabah: AI-Assisted Cocoa Bean Quality Screening & Preliminary Assessment Portal is a web app that combines an AI check of a bean photo, manually entered measurements and simple quality rules into a preliminary assessment with a recommended next action.

- **Team:** Synergy IT Team, 5 members, led by Firdaus Rusdianshan
- **Programme:** Gamuda AI Academy Sabah, Cohort 5 (7 Sept – 3 Dec 2026); final presentation on Demo Day in week 12
- **Next deadline:** User Requirement Document (URD) due 9 October
- **Goal:** a working proof-of-concept for the capstone, not a production system

What it is and is not

CocoaCheck is a **decision-support and preliminary screening** prototype. The person using it stays responsible for the final decision.

It is **not**:

- an official cocoa grading system
- a replacement for professional inspectors
- a moisture meter or lab test
- able to judge taste, aroma or fermentation from a photo
- a price calculator or sorting machine

Problem statement

The problem is still a **hypothesis**: cocoa quality assessment involves several separate checks, measurements and judgement, and parts of it could be supported by one structured digital workflow. We will validate this with **Lembaga Koko Malaysia (LKM)**. Until then, we do not claim that current assessment is slow, inaccurate or lacks a system.

The AI part (working assumption)

The AI looks at a photo of **one whole dried cocoa bean** and says **normal** or **non-normal**, with how confident it is. It learns from an existing set of labelled bean photos, so we don't need to collect and label our own.

Pod vs bean: the clarification

	Cocoa pod (fruit)	Whole dried bean	Cut-test bean
Looks like	Large fruit, usually on the tree	Small brown seed after drying	Bean cut in half to show the inside
Typical labels	Plant diseases (black pod, moniliasis)	Physical defects (broken, flat, mouldy)	Inside colour (brown, purple, slaty)
Used in CocoaCheck?	No	**Yes (MVP)**	Future enhancement

Dataset

- **Source:** cacao classification on Roboflow Universe
- **Size:** 694 images, one whole dried bean per image
- **Original classes:** good, broken, insect, cluster, flat, germinated, moldy
- **Licence:** CC BY 4.0, so we must credit the source in the report and slides

We merge the classes into two:

Our class	Original classes
normal	good
non_normal	broken, insect, cluster, flat, germinated, moldy

Backup plan

If the main AI tool doesn't work out, we switch to **Google's Gemini AI** instead. The rest of the system stays the same.

Known limitations

- The dataset is not from Sabah.
- The accuracy shown by Roboflow is on its own test images; on our phone photos it will likely be lower, so we test early.
- The AI cannot see moisture, fermentation or internal defects.

Workflow (Phase 1, no login)

The user goes through eight steps in order. AI is only one of them.

1. **Create assessment**: batch ID, source/supplier, assessment date, notes
2. **Upload bean photo**: one whole bean from the batch, on a plain background with good lighting. The system checks file type and size and shows a preview.
3. **AI screening**: the AI checks the photo and says normal or non-normal, with how confident it is
4. **Enter physical measurements** (typed in by the user):
   - sample weight (g)
   - total bean count
   - moisture (%)
   - foreign matter (%), if applicable
   - notes
5. **Rule check**: the rule engine compares the AI result and measurements against the rules
6. **Preliminary result**: result, reasons, recommended next action
7. **Review**: the user checks everything in one place and can go back to edit before confirming
8. **Generate report**: a PDF with batch info, photo, AI result, measurements, result, recommendation and a disclaimer that it is not official grading

About the single photo (working assumption)

One bean photo stands in for the whole batch. This is simple, but it is the weakest point of the design, because one bean cannot statistically represent a batch. To stay honest:

- Call it a **spot check** of a sampled bean.
- Say "the sampled bean appears non-normal", never "the batch is non-normal".

A possible upgrade, to ask the trainer about: allow 3–5 bean photos per assessment and report "2 of 5 sampled beans non-normal".

Rule engine (demo values)

A **rule** is an "if… then…" check, and a **threshold** is the limit it compares against. Example: "moisture must be at most 7.5%"; if the user enters 8.2%, the rule fails. The rule engine runs every rule and combines the results.

Until LKM or the MS 293 standard gives real values, we use **demo thresholds**, labelled as DEMO on the screen, the report and the slides.

#	Rule	Threshold (DEMO)	If it fails
R1	AI result is normal	—	"Visible abnormality in sampled bean"
R2	AI confidence is high enough	at least 70%	"AI uncertain, manual check needed"
R3	Moisture is not too high	at most 7.5% (to verify)	"Moisture above limit"
R4	Bean count is not too high	at most 100 beans per 100 g	"High bean count"
R5	Foreign matter is low	at most 1%	"Foreign matter above limit"

Possible results

- No rule failed → **Passed preliminary screening**
- Any rule failed → **Further inspection recommended**, with the reasons listed
- Only R2 failed → **Needs manual review**

What we build first

Phase	Contents
1 – Core (must work)	Create assessment, photo upload, AI, measurements, rule engine, result, report
2 – Login	Login, Staff/Admin roles, link assessments to users
3 – Admin	Admin dashboard, view assessments and who did them, view reports
4 – Management	Staff management, rule management, activity logs
5 – Future	Cut-test AI, more advanced AI

Phases 2–5 are built only if time allows.

Demo Day, scope limits and open decisions

Demo Day plan (decided: live demo)

- Bring two batches of dried beans: one clearly normal, one clearly non-normal.
- Photograph one bean from each batch live and run the full workflow to the PDF report.
- Say openly that these are clear examples; borderline beans are where human review matters.
- Keep backup photos of both batches on the laptop.
- Bring a mobile hotspot in case the venue Wi-Fi fails.
- Run one assessment 5 minutes before presenting, so the system is ready.
- Test with our own phone photos well before Demo Day.

Not in the MVP

- Login, registration, admin panel, user and role management, activity logs
- Cut-test AI
- Official grading or fair-price calculation
- Moisture or fermentation from photos
- IoT, sensors or sorting hardware

Claims we must avoid

- "The AI grades cocoa" or "replaces inspectors"
- "The system makes assessment faster" (not measured)
- "Current assessment is slow or inaccurate" (not validated)
- "The batch is non-normal" based on one bean
- Presenting demo thresholds as official standards

Open decisions

Question	Who decides	When
Is our AI approach (existing tool + public dataset) acceptable?	Trainer	5 Oct
One bean photo per assessment, or 3–5?	Trainer + team	5 Oct
Team roles	Team	Before URD
LKM interview date	LKM	Pending

Question	Who decides	When
Real quality rule values	LKM	After interview
Show login/admin as "planned" in the URD, or leave them out?	Team	Before URD (9 Oct)

Sources

- cacao classification dataset (Roboflow Universe), CC BY 4.0